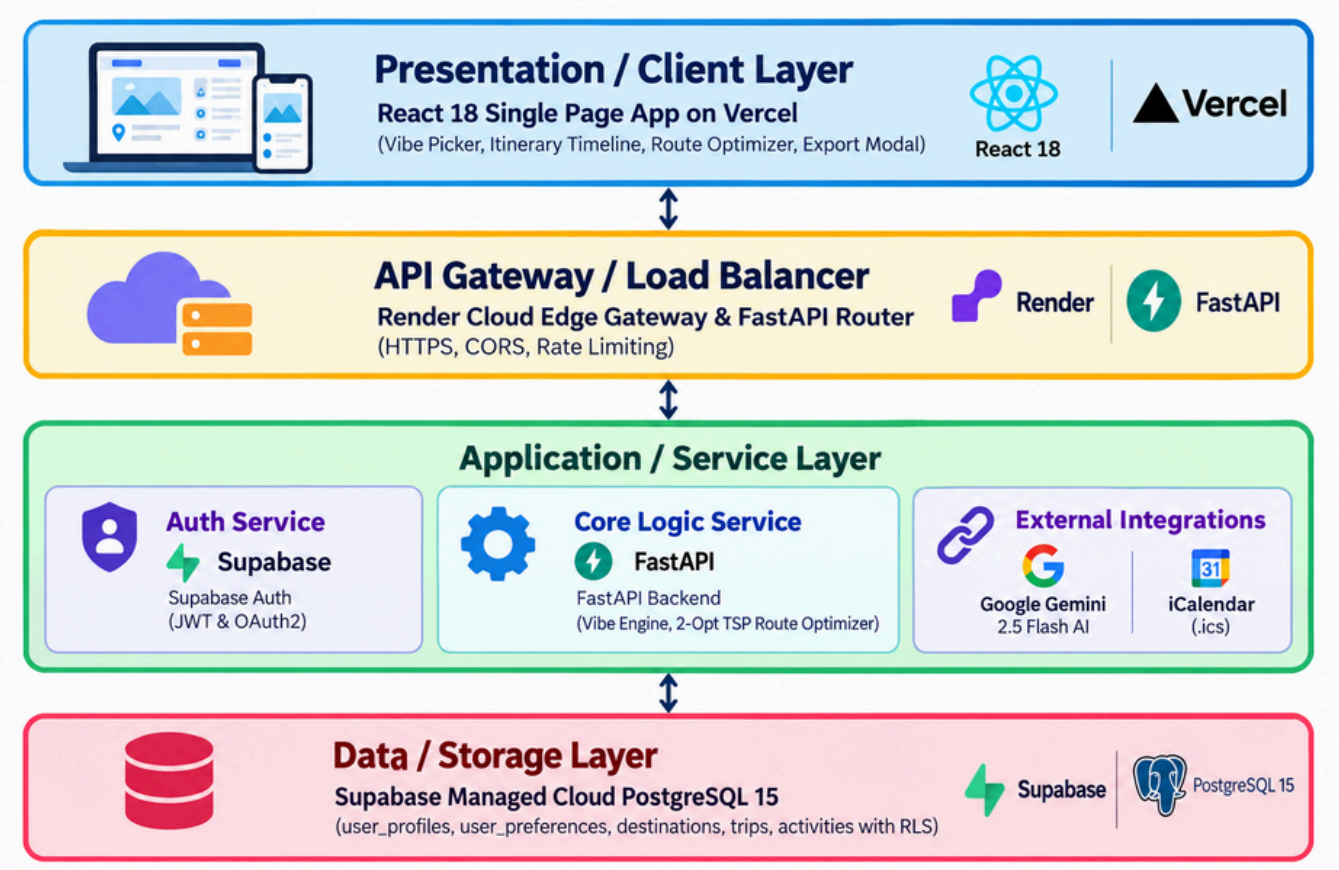
Solution Architecture

Date	16 October 2026
Team ID	
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-levelvisual representation of your solution's architecture. Use the structural ayout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	Renders the responsive glassmorphism UI, captures traveler vibes, displays daily timelines, and handles client-side state.	React 18, Vite, JavaScript, CSS3 (Hosted on Vercel)
[API Gateway]	Serves as the single entry point for client requests, directing REST endpoints, managing CORS policies, and health monitoring.	FastAPI Router & Render Edge Proxy
[Microservices]	Executes core business logic: prompt orchestration for Gemini AI, 2-opt TSP route sorting, and .ics/PDF export formatting.	Python 3.11, FastAPI, Uvicorn, Google GenAI SDK, Haversine
[Database]	Provides persistent cloud relational storage for trips, user preference vectors, and destination seed catalogs with RLS security.	Supabase (PostgreSQL 15)